MODULE 03

Virtual Machines and Networks in the Cloud

Running workloads on Compute Engine and connecting them securely – to each other, to users, and to your own data centers



What we will cover today

01
VPC networking
Your private network inside Google Cloud

02
Compute Engine
Virtual machines, images, pricing and storage

03
Scaling VMs
Scaling out vs scaling up

04
VPC capabilities
Routes, firewalls, peering and Shared VPC

05
Cloud Load Balancing
Spreading traffic across instances

06
Cloud DNS
Turning names into addresses

07
Cloud CDN
Caching content close to users

08
Connecting networks
VPN, peering and interconnect options

By the end of this module you will be able to

- Explain what a VPC is and why its networks are global
- Describe Compute Engine VMs, images and pricing options
- Choose between scaling out and scaling up
- Use firewall rules, routes, peering and Shared VPC
- Explain how load balancing, DNS and CDN serve users
- Pick the right way to connect other networks to a VPC



Module format

8 topics
12-question knowledge check
2 hands-on lab

01

VPC networking



A private, isolated network for your resources – hosted on Google's public cloud

A VPC is a private cloud inside a public one



Private cloud

Isolated and under your control, but you buy and run the hardware

+



Public cloud

Massive scale and convenience, shared infrastructure

=



Virtual Private Cloud

Public-cloud scale with private-cloud isolation



Think of a private, locked office suite inside a huge shared tower: the building provides power, security and lifts – your suite is yours alone. Inside it you can run code, store data and host websites.

What a VPC network does for you

Connects resources

Links your Google Cloud resources to each other and to the internet

Segments the network

Divides the network into subnets for different workloads

Restricts access

Firewall rules control which traffic reaches instances

Directs traffic

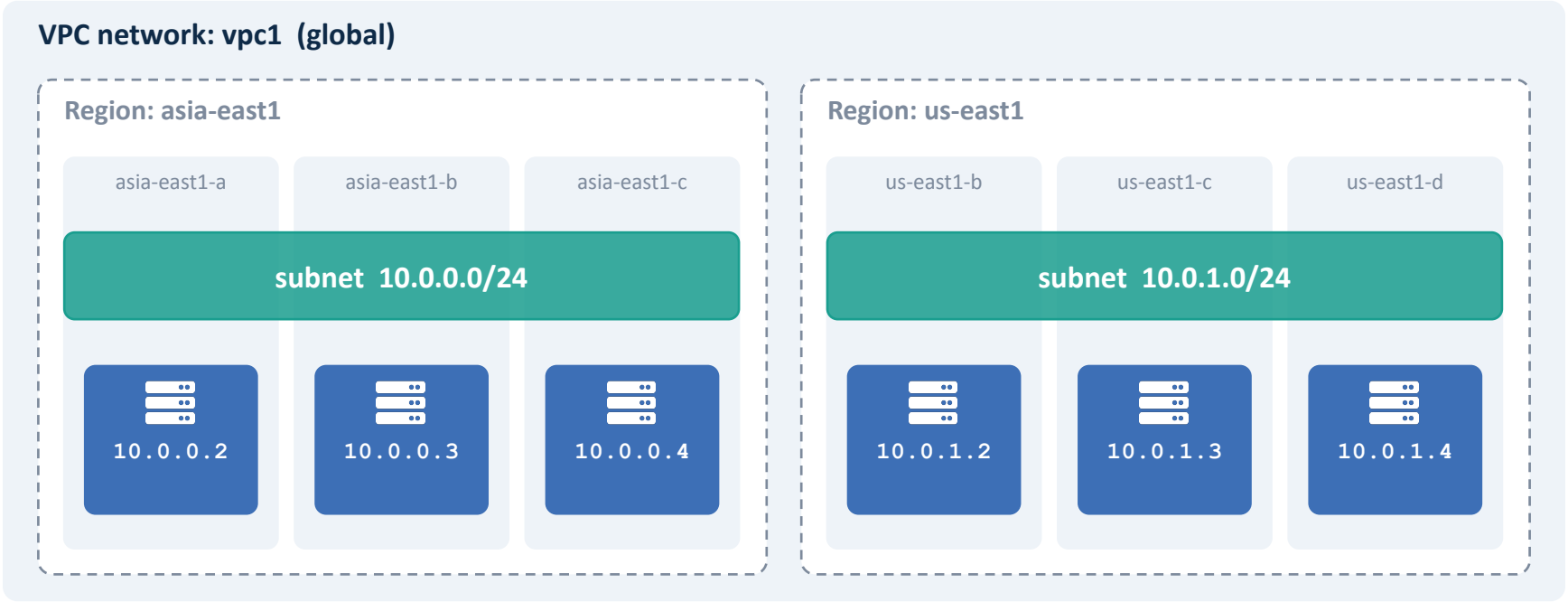
Static routes send traffic to specific destinations



The surprise

VPC networks are global. One network can have subnets in any Google Cloud region worldwide.

Subnets are regional and span zones



Three VMs in three different zones are still neighbors on the same subnet – resilient, yet simple.

02

Compute Engine



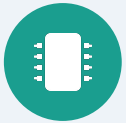
Google Cloud's Infrastructure as a Service: virtual machines running on Google's infrastructure

A VM you configure like a physical server

- Create and run virtual machines on Google's infrastructure
- No upfront investment – run thousands of virtual CPUs
- Built for fast, consistent performance
- Each VM has the power of a full operating system

Same decisions as buying a server – but made in seconds and changeable later.

YOU CHOOSE



vCPUs

Processing power



Memory

RAM for your workload



Storage

Amount and type of disk



Operating system

Linux, Windows or custom

Many ways to create a VM, many images to run

HOW YOU CREATE IT

Google Cloud console

Web-based point and click

Google Cloud CLI

```
gcloud compute instances create
```

Compute Engine API

Automate from your own code

WHAT IT RUNS



Linux images

Provided by Google



Windows Server images

Provided by Google



Customized versions

Your own tailored images



Other operating systems

Build and run your own

Cloud Marketplace: ready-made solutions



Cloud Marketplace

Pre-configured software from Google and third-party vendors, deployed in a few clicks

No manual setup

Software, VMs, storage and network settings come pre-configured – and many can be adjusted before launch

Usually no extra charge

Most packages cost nothing beyond normal Google Cloud resource fees

Licensed software may cost more

Some third-party images charge usage fees, but every solution shows a monthly cost estimate before launch

Pricing options that reward the right usage

Per second

billing

One-minute minimum, then pay by the second

Automatic

sustained-use discount

Kicks in for VMs running more than 25% of a month – no action needed

Up to 57%

committed-use discount

Commit to vCPUs and memory for 1 or 3 years for stable, predictable workloads

Up to 90%

off with Spot VMs

For fault-tolerant jobs; Google can reclaim them when capacity is needed

Spot VMs have no maximum runtime, but design jobs to stop and restart without losing progress.

Pay only for the storage and size you need

Storage option	What it is	Good for
Zonal persistent disk	Efficient, reliable block storage	The common default for VMs
Regional persistent disk	Block storage replicated across two zones	Surviving a zone failure
Local SSD	High-performance, transient local block storage	Scratch space, caches
Cloud Storage bucket	Affordable object storage	Files, backups, media
Filestore	High-performance file storage	Shared file systems

Not sure? Start with a persistent disk. High throughput between VMs and disks is included at no extra cost.



Custom machine types

Choose a predefined machine type, or set your own number of vCPUs and amount of memory – so you never pay for capacity you don't use.

Up next: hands-on lab



Create Virtual Machines

Compute Engine gives you a lot of choice when you build a virtual machine: the machine size, the operating system, the disk, the network settings and how the VM behaves during maintenance. In this lab you build three VMs, each with a different purpose, and look at which settings you can change after creation and which are fixed.

1. How to create a standard Compute Engine VM and compare machine types and costs
2. How to create a Windows VM and prepare it for a Remote Desktop connection
3. How to create a Windows VM and prepare it for a Remote Desktop connection
4. Verify Instances

03

Scaling virtual machines



Matching capacity to demand – by adding more VMs, or by making one VM bigger

Scale out or scale up?

Scale out

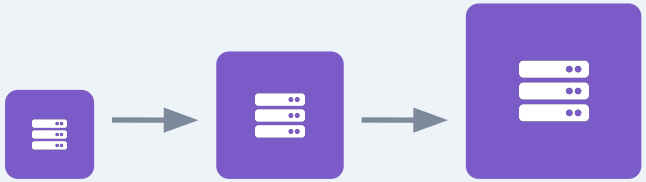
Autoscaling: add or remove VMs



- VMs are added or removed based on load metrics
- Pair with load balancing to spread traffic
- Where most customers start

Scale up

Use a bigger VM



- Very large VMs suit in-memory databases and CPU-heavy analytics
- Max vCPUs depends on machine family
- Also limited by your zone's quota

Machine types: pick a family, then a size



General purpose

Balanced price and performance for most workloads



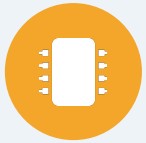
Compute-optimized

Highest performance per core for compute-heavy work



Memory-optimized

Very large memory for in-memory databases and analytics



Accelerator-optimized

GPUs for machine learning and high-performance computing



Current specs for every machine type: cloud.google.com/compute/docs/machine-types

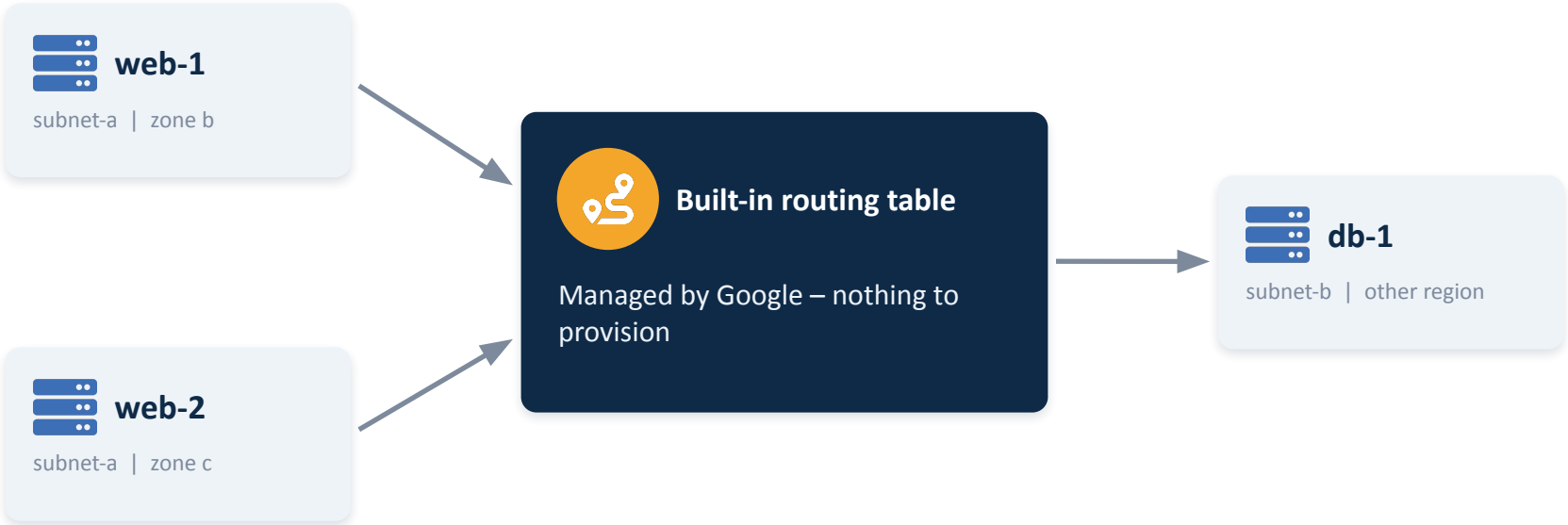
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VPC capabilities



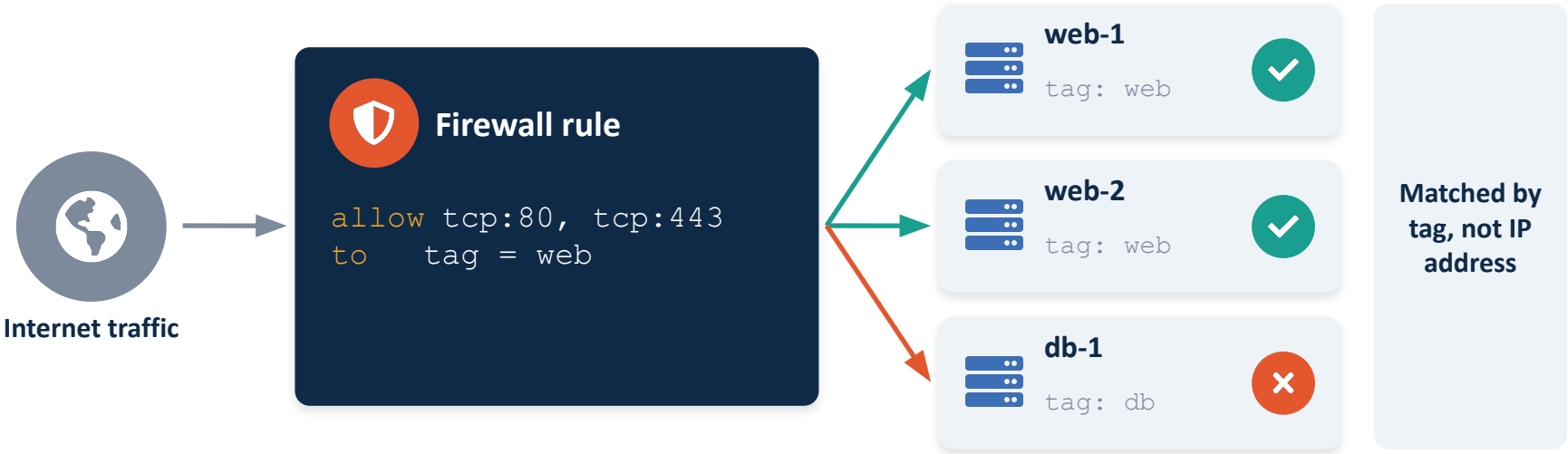
Routing, firewalls and cross-project networking – built in, with nothing to rack or patch

Routing is built in – no router to manage



Forwards traffic within a network, across subnets and across zones – no external IP address needed.

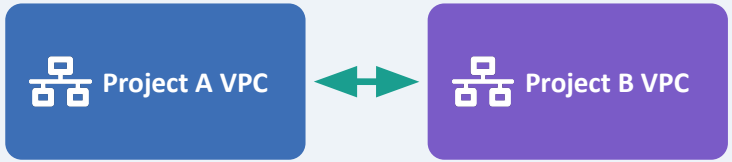
A distributed firewall, controlled with tags



- Global and distributed – nothing to provision
- Controls both incoming and outgoing traffic
- New VMs tagged "web" are covered automatically

Connecting VPCs across projects

VPC Network Peering



- A relationship between two VPCs so they can exchange traffic
- Each side keeps managing its own network

Shared VPC

One network in a host project, used by service projects

- Uses the full power of IAM to control who and what in one project can use a VPC in another
- Central control by a network team

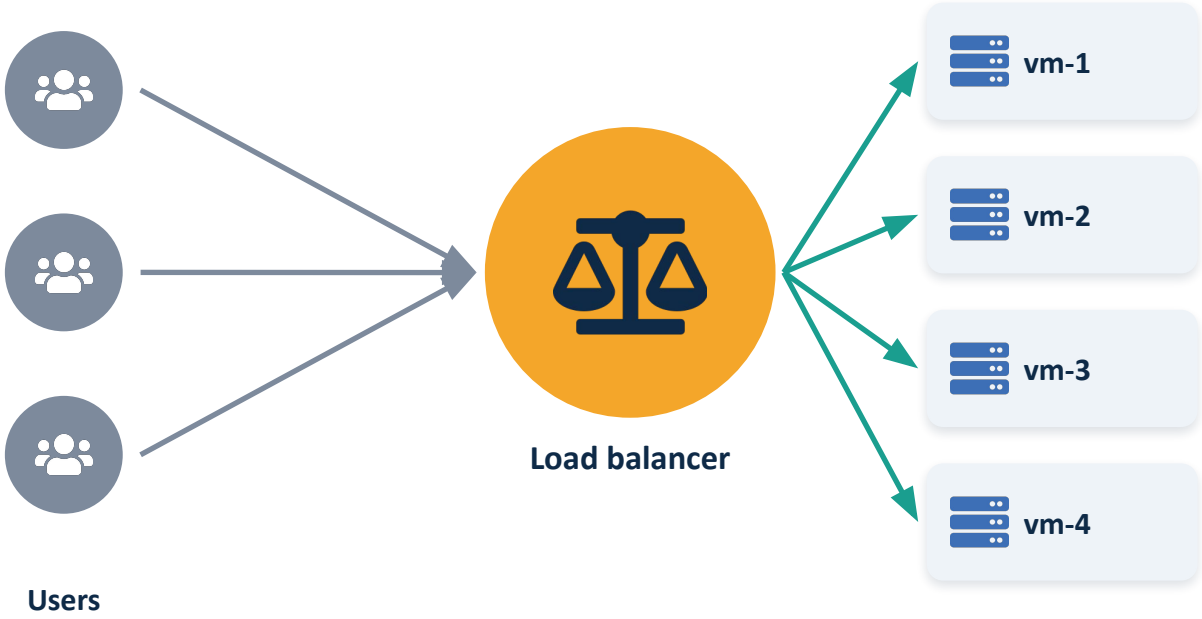
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Cloud Load Balancing



Giving users one front door to an application that may run on 4 VMs one minute and 40 the next

One front door, many backends



Why it matters

- Spreads traffic across instances
- Reduces the risk of slowdowns
- Users never need to know how many VMs exist

What makes Cloud Load Balancing different



Fully managed

Software-defined and distributed – it doesn't run on VMs you have to scale or patch



All your traffic

HTTP(S), TCP, SSL and UDP traffic



Single or cross-region

Includes automatic multi-region failover, shifting traffic gradually away from unhealthy backends



No pre-warming

Handles sudden spikes without a support ticket or advance notice

Choosing a load balancer



Application Load Balancer

Layer 7 | HTTP and HTTPS

- Acts as a reverse proxy
- Content-based routing and SSL/TLS termination
- External or internal; global, regional or cross-region

Best for web apps and APIs



Network Load Balancer

Layer 4 | TCP, UDP, other IP protocols

Proxy

Terminates client connections and opens new ones to backends; supports on-premises and multi-cloud backends

Passthrough

Forwards traffic untouched, preserving the client's source IP

06

Cloud DNS



Helping the world find your applications by name

From names to addresses



You type

`shop.example.com`



DNS looks it up

Like a phone book for the internet



You reach

`203.0.113.10`

8.8.8.8

Google's free public DNS, used worldwide



Cloud DNS

- Managed DNS on the same infrastructure as Google; low latency, high availability, cost-effective
- Served from redundant locations worldwide
- Programmable: manage millions of zones and records via console, CLI or API

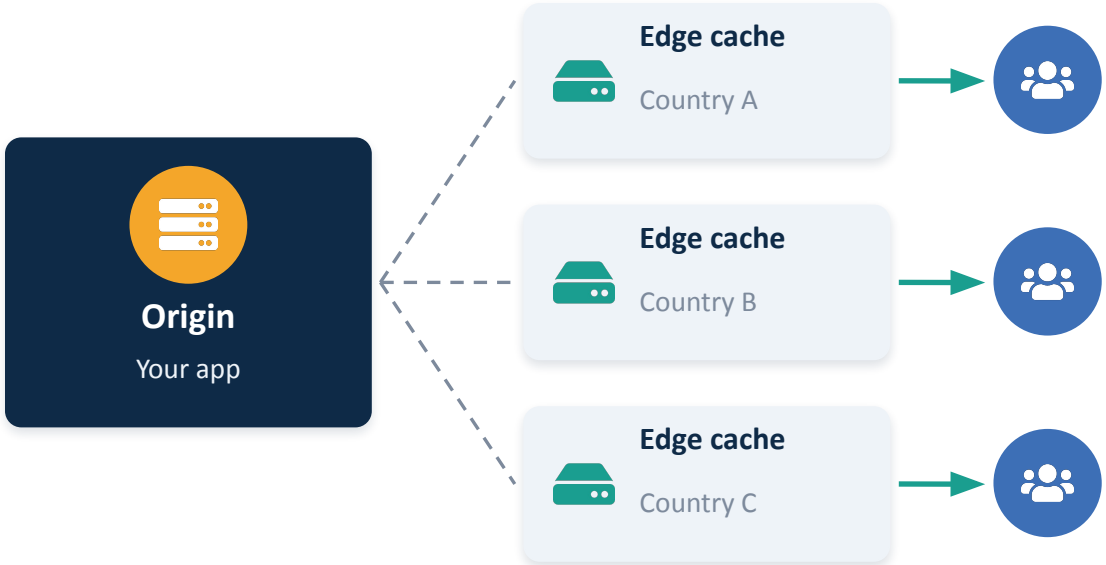
07

Cloud CDN



Using Google's global edge caches to serve content close to your users

Cache content at the edge, closer to users



Dashed lines: content is fetched from the origin once, then served from the edge.

Cloud CDN benefits

- Lower latency for users
- Less load on your origin
- Can save money
- Enabled with one checkbox

08

Connecting networks to a VPC

Linking Google Cloud with your on-premises data centers and other clouds



Six ways to connect, in three families

Over the internet

Cloud VPN

Encrypted tunnel

Peering

Direct Peering

Your router at a Google PoP

Carrier Peering

Through a service provider

Interconnect

Dedicated Interconnect

Private direct connection

Partner Interconnect

Through a supported provider

Cross-Cloud Interconnect

To another cloud provider

Cloud VPN: a secure tunnel over the internet



On-premises network



Encrypted VPN tunnel over the internet

BGP route exchange



Cloud Router

inside your Google VPC



Dynamic routing

Cloud Router exchanges routes with your network using Border Gateway Protocol (BGP)



New subnets just appear

Add a subnet in Google Cloud and your on-premises network learns the route automatically



Internet trade-offs

Not ideal when security policy or bandwidth reliability rules out using the internet

Peering: connect at Google's network edge



Direct Peering

- Put your own router in a public data center alongside a Google point of presence (PoP)
- Exchange traffic directly between networks
- Google has more than 200 PoPs worldwide

Best when: You already have, or will lease, space at a Google PoP



Carrier Peering

- A partner in the Carrier Peering program connects you
- Direct access from on-premises, through the provider's network
- Reaches Google Workspace and Google Cloud services with public IPs

Best when: You aren't in a PoP and prefer a provider



Peering is not covered by a Google Service Level Agreement (SLA).

Interconnect: private, high-bandwidth links

Dedicated Interconnect

- One or more direct, private connections to Google
- Highest uptime option
- Can be backed up by a VPN

Up to 99.99% SLA*

Partner Interconnect

- Connects through a supported service provider
- For sites that can't reach a colocation facility, or don't need a full 10 Gbps
- Configurable for mission-critical or downtime-tolerant apps

Up to 99.99% SLA*

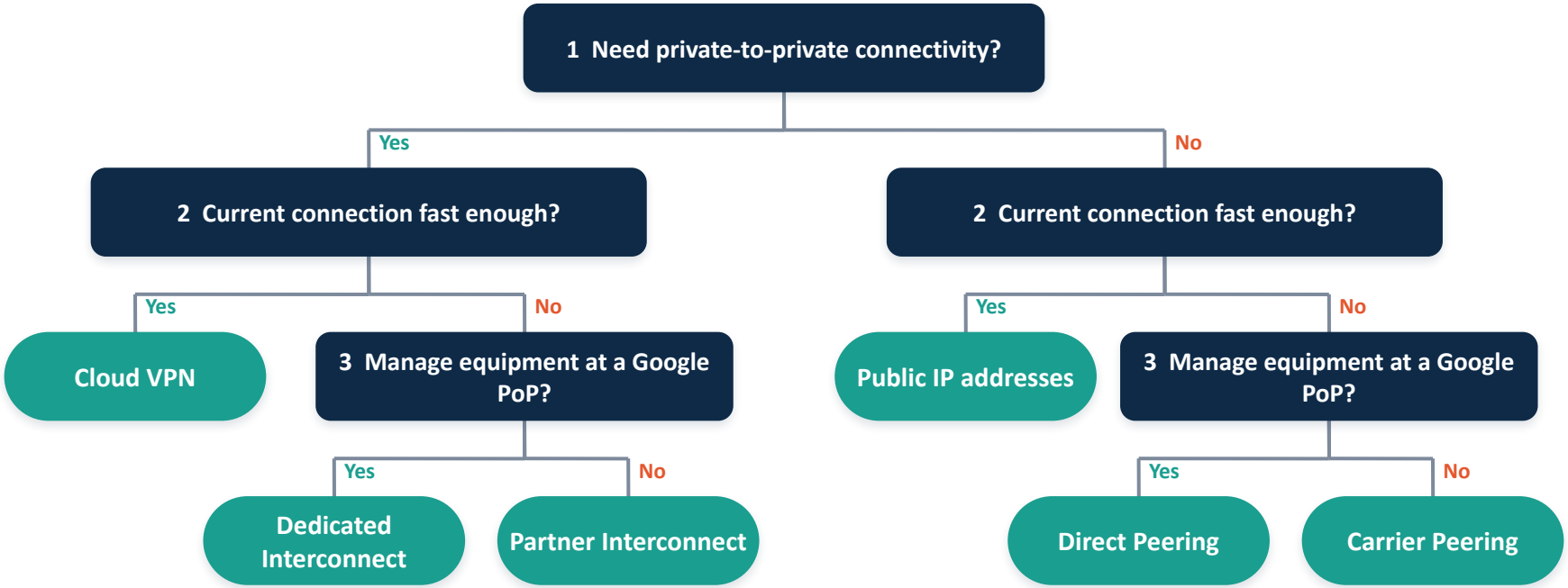
Cross-Cloud Interconnect

- Dedicated physical link from Google to another cloud provider
- Supports an integrated multicloud strategy
- Encrypted, site-to-site transfer

10 Gbps or 100 Gbps

** When connection topologies meet Google's specifications. For Partner Interconnect, Google isn't responsible for the provider's part.*

Which connection should you choose?



Dark boxes are questions; teal boxes are recommendations. Follow Yes or No at each split.

Match the need to the service

SCENARIO	BEST FIT
A batch job crunches data overnight and can be restarted	**Spot VMs**
Only VMs tagged "web" should accept HTTPS traffic	**Firewall rule + tag**
Images and videos load slowly for overseas users	**Cloud CDN**
A central team should govern one network for 10 projects	**Shared VPC**
A data center needs private links but can't reach a colocation site	**Partner Interconnect**

Key takeaways

1

VPC

Global networks with regional subnets that span zones

2

Compute Engine

Configurable VMs; save with sustained-use, committed-use and Spot

3

Scaling

Scale out with autoscaling; scale up with bigger machine types

4

Built-in network

Routing and a tag-based firewall – nothing to provision

5

Serving users

Load balancing, Cloud DNS and Cloud CDN

6

Hybrid

VPN, peering or interconnect – choose with three questions

?

Knowledge check



12 questions covering all eight topics. Answer each one before we reveal it.

Question 1 of 6

VPC networking

What is the scope of a Google Cloud VPC network?

A Zonal

B Regional

C Global

D Limited to one data center



Hint (if needed): One network can have subnets in any region worldwide.

Answer 1: C

VPC networking

What is the scope of a Google Cloud VPC network?

A Zonal

B Regional

✓ **Global**

D Limited to one data center

Why

VPC networks are global. A single VPC can contain subnets in any Google Cloud region.

Question 2 of 6

Subnets

Three VMs in us-east1-b, us-east1-c and us-east1-d share one subnet. Why is that possible?

A. Subnets are zonal, so each zone shares it

B. Subnets are regional and span all zones in a region

C. The VMs must have external IP addresses

D. It isn't possible; each zone needs its own subnet



Hint (if needed): What is a subnet's scope?

Answer 2: B

Subnets

Three VMs in us-east1-b, us-east1-c and us-east1-d share one subnet. Why is that possible?

A Subnets are zonal, so each zone shares it

✓ **Subnets are regional and span all zones in a region**

C The VMs must have external IP addresses

D It isn't possible; each zone needs its own subnet

Why

Subnets are regional and stretch across the zones in their region, so VMs in different zones can be neighbors on one subnet.

Question 3 of 6

Pricing

A nightly batch job can be stopped and restarted without losing progress. Which option cuts cost the most?

A. Committed-use discount

B. Spot VMs

C. Regional persistent disk

D. A larger machine type



Hint (if needed): Google may reclaim these VMs, but the discount is big.

Answer 3: B

Pricing

A nightly batch job can be stopped and restarted without losing progress. Which option cuts cost the most?

A Committed-use discount

Spot VMs

C Regional persistent disk

D A larger machine type

Why

Spot VMs offer up to 90% off standard pricing for fault-tolerant workloads; Google can reclaim them when it needs the capacity.

Question 4 of 6

Pricing

A database runs 24/7 and will for the next three years. Which pricing option fits best?

A Spot VMs

B Committed-use discount

C Per-second billing only

D Cloud Marketplace



Hint (if needed): Stable and predictable usage can be committed in advance.

Answer 4: B

Pricing

A database runs 24/7 and will for the next three years. Which pricing option fits best?

A Spot VMs	✓ **Committed-use discount**
C Per-second billing only	D Cloud Marketplace

Why

Committed-use discounts give up to 57% off in return for a 1- or 3-year commitment to vCPUs and memory – ideal for steady workloads.

Question 5 of 6

Pricing

When do sustained-use discounts start to apply?

A Only after you sign a one-year contract

B Automatically once a VM runs for more than 25% of a month

C Only for Spot VMs

D Only when you request them from support



Hint (if needed): No action is needed to get them.

Answer 5: B

Pricing

When do sustained-use discounts start to apply?

A Only after you sign a one-year contract

✓ **Automatically once a VM runs for more than 25% of a month**

C Only for Spot VMs

D Only when you request them from support

✓ **Why**

Sustained-use discounts apply automatically: once a VM runs for more than 25% of a month, every additional minute is discounted.

Question 6 of 6

Storage

Which storage option gives high-performance but transient block storage attached to the VM?

A Local SSD

B Regional persistent disk

C Cloud Storage bucket

D Filestore



Hint (if needed): Fast and close, but not built to last.

Answer 6: A

Storage

Which storage option gives high-performance but transient block storage attached to the VM?

- ✓ **Local SSD**
- B Regional persistent disk
- C Cloud Storage bucket
- D Filestore

Why

Local SSD is high-performance, transient, local block storage – great for scratch data and caches, not for data you must keep.

Up next: hands-on lab



Get started with VPC networking and Compute Engine

Build an auto mode VPC network with firewall rules and two VM instances, then test how the VMs connect.

1. Create an auto mode VPC network
2. Add firewall rules for the network
3. Create two VM instances in different zones
4. Test connectivity between the VMs

THANK YOU

Questions and discussion



VPC | Compute Engine | Scaling | Firewalls | Load balancing | DNS | CDN | Hybrid connectivity